

B.TECH. COMPUTER SCIENCE AND ENGINEERING (AI& ML)
Deenbandhu Chhotu Ram University of Science & Technology Murthal, Sonapat, Haryana
Semester – III (Effective from Session 2025-26)

Course Code: PCCSE205	Minor	TCA	TES	PCA	PES	Total
Title of the Course: DATABASE MANAGEMENT SYSTEMS	15	10	50	15	10	100
L-T-P: 3-0-2						
Credits: 4	Duration of Exam: 3 Hours					
Brief Write-Up About the Course (100-150 words) This course introduces the foundational concepts and techniques of database systems, focusing on relational data models and structured query languages. It begins with data modeling using Entity-Relationship (ER) diagrams and progresses to relational algebra and relational calculus. Students learn SQL for data definition, manipulation, and advanced query handling including nested queries and aggregations, along with an introduction to NoSQL systems like MongoDB. The course also covers database design principles, functional dependencies, and normalization up to 5NF. It explores transaction management, concurrency control mechanisms, and recovery techniques ensuring data integrity and ACID compliance. Finally, students gain insights into data storage methods and indexing structures such as B+ trees and hashing. By the end of the course, students will be able to design efficient relational databases and execute complex queries with an understanding of underlying storage and transaction protocols.						
Course Objective (CO): 1. To understand basic database concepts, including the structure and operation of the relational data model. 2. To construct simple and moderately advanced database queries using Structured Query Language. 3. To apply logical database design principles with database normalization. 4. To understand the concept of a database transaction, concurrency control, recovery, and Indexing techniques.						
UNIT I Introduction And Relational Data Models: Introduction to database systems, Various components of a DBMS; ER Model - Conceptual data modeling - entities, entity types, various types of attributes, relationships, relationship types, ER diagram, Concept of relations, schema-instance distinction, keys, referential integrity and foreign keys, relational algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relation calculus, domain relational calculus, converting the database specification in ER notation to the relational schema.						
UNIT II Structured Query Language: Introduction, data definition in SQL, table, key and foreign key definitions, update behaviours. Querying in SQL - basic select-from where block and its semantics, nested queries - correlated and uncorrelated, notion of aggregation, aggregation functions group by and having clauses, embedded SQL, Introduction to NoSQL and MongoDB.						
UNIT III Dependencies and Normal Forms: Various Database Design Strategies, Functional Dependencies, dependency theory - functional dependencies, Armstrong's axioms for FD's, closure of a set of FD's & its applications (equivalences & canonical form), Normalization for						

Relational Databases: motivation for normal forms, definitions of 1NF, 2NF, 3NF and BCNF, decompositions and desirable properties of them, algorithms for 3NF and BCNF normalization, multi-valued dependencies and 4NF, join dependencies and definition of 5NF.

UNIT IV

Transaction Processing, Error Recovery: Transaction Processing: concepts of transaction processing, ACID property, isolation problems, schedules and recoverability, serializability of schedules. Concurrency Control: Locking based protocol, Time-stamp based protocol, multi-version schemes, Validation based protocol, Multiple granularity. Error Recovery: Failure classification, deferred update, immediate update, Shadow paging.

Text books:

1. Database system concepts, Korth & Sudarshan, MH.
2. Introduction to Database Systems, C.J.Date, Pearson Education.

Reference Books:

1. Principles of Database Systems, 2nd Edn., Ullman, J.O, Galgotia Publications.
2. Fundamentals of Database Systems, Elmasri & Navathe, Pearson Education. Database Design Fundamentals, Rishe, PHI.
3. Database Management Systems, Ramakrishnan & Gehrke, MH.

Course Outcome (CO):

After successful completion of the course, students will be able:

1. To implement basic database concepts, including the structure and operation of the relational data model.
2. To construct simple and moderately advanced database queries using Structured Query Language (SQL) and solve simple queries.
3. To apply logical database design principles with database normalization.
4. To understand the concept of a database transaction, concurrency control, recovery, and Indexing techniques.

List of Programs:

1. To study Data Definition language
 - 1.1. Create, alter, drop, truncate
 - 1.2. To implement Constraints.
 - 1.2.1.(a). Primary key, (b). Foreign Key, (c). Check, (d). Unique, (e). Null, (f). Not null, (g). Default, (h). Enable Constraints, (i). Disable Constraints (j). Drop Constraints
2. To implementation on DML, TCL and DRL
 - 2.1. (a). Insert, (b). Select, (c). Update, (d). Delete, (e). commit, (f). rollback, (g). save point, (h). Like '%', (i). Relational Operator (j) Logical operators
3. To implement Nested Queries & Join Queries
 - 3.1. To implementation of Nested Queries
 - 3.2. (a) Inner join, (b). Left join, (c). Right join (d). Full join (e) Natural Join (f) Theta Join (g) Cross Join
4. To implement Views
 - 4.1. (a). View, (b). joint view, (c). force view, (d). View with check option
5. To implement Index
 - 5.1. (a) Normal Index (b) Unique Index (c) Bitmap Index (d) Composite (e) B-Tree (f) Clustered (g) Non- Clustered
6. To study various Date handling functions and their usage.
7. To Study String Handling Functions with Usage.
8. Case study of oracle 11g with emphasis on Recovery Techniques, Concurrency and Transaction Management.